

1. Write the I.U.P.A.C name of $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$.

Ans: 1-Propanamine **OR** Propanamine

2. Which of the following groups when present at para position increases the basic strength of aniline?

- (a) $-\text{NO}_2$ (b) $-\text{Br}$ (c) $-\text{NH}_2$ (d) $-\text{COOH}$

3. The reagent used in Sandmeyer's reaction is

- (a) N_2 gas (b) Cu and HCl (c) Cu_2Cl_2 and HCl (d) CuCl_2

4. Consider the given reaction: $\text{RCONH}_2 + \text{Br}_2 + 4 \text{NaOH} \longrightarrow \text{RNH}_2 + \text{Na}_2\text{CO}_3 + 2 \text{NaBr} + 2 \text{H}_2\text{O}$

5. If the molecular formula of an amine is CH_5N , then the molecular formula of amide will be

- (a) $\text{C}_2\text{H}_6\text{NO}$ (b) $\text{C}_2\text{H}_5\text{NO}$ (c) CH_5NO (d) $\text{C}_2\text{H}_4\text{NO}$

6. The chemical formula of Hinsberg's reagent is

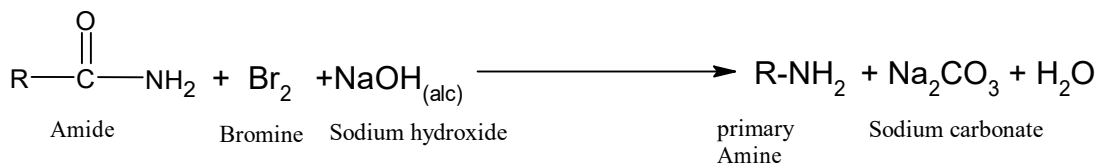
- (a) $\text{C}_6\text{H}_6\text{SOCl}$ (b) $\text{C}_6\text{H}_5\text{SOCl}_2$ (c) $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$ (d) $\text{C}_6\text{H}_5\text{SO}_3\text{H}$

7. Gabriel Synthesis is used for the preparation of _____ amines (**Primary**)

8. The colours of p-Hydroxyazobenzene is _____ (**Orange**)

9. Explain Hoffmann bromamide degradation reaction with chemical equation.

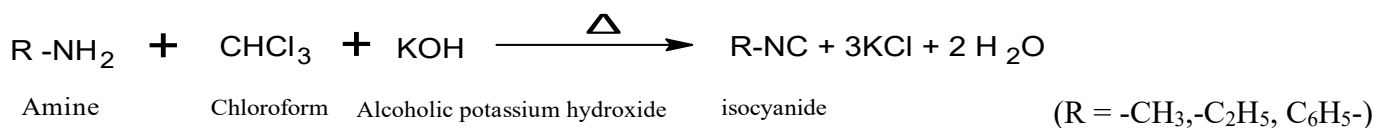
Ans: treating an amide with bromine in an aqueous or ethanolic solution of sodium hydroxide to form primary amine



(**R = $-\text{CH}_3$ OR C_2H_5 - OR C_6H_5 -**)

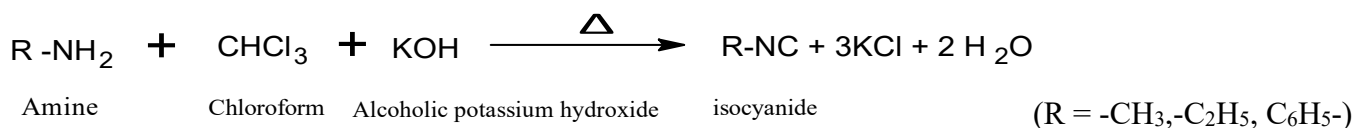
1. Which type of amines answers for carbylamine reaction (isocyanide test)? Explain this reaction with an example.

Ans: Aliphatic and aromatic primary amines

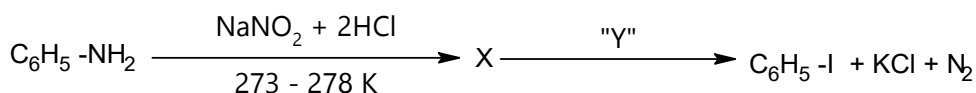


11. Explain Carbylamine reaction with equation

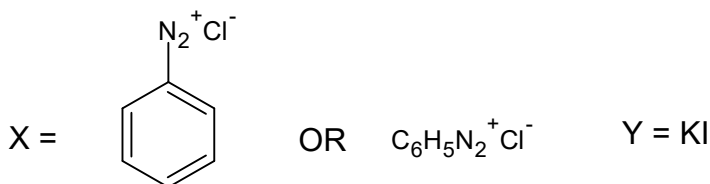
Ans: Aliphatic and aromatic primary amines on heating with chloroform and ethanolic potassium hydroxide form isocyanides or carbylamines



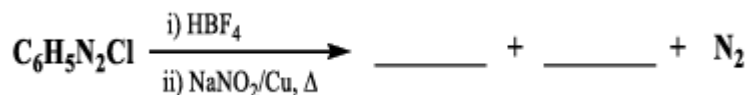
12. Identify the compounds "X" and "Y" in the following conversion



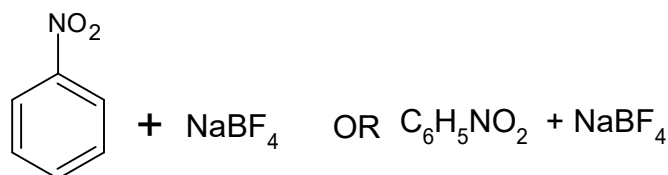
Ans:



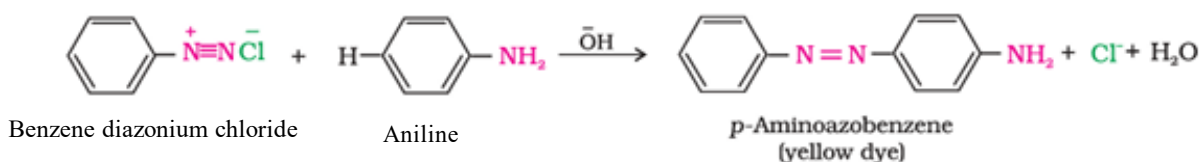
13. Complete the following reaction:



Ans:



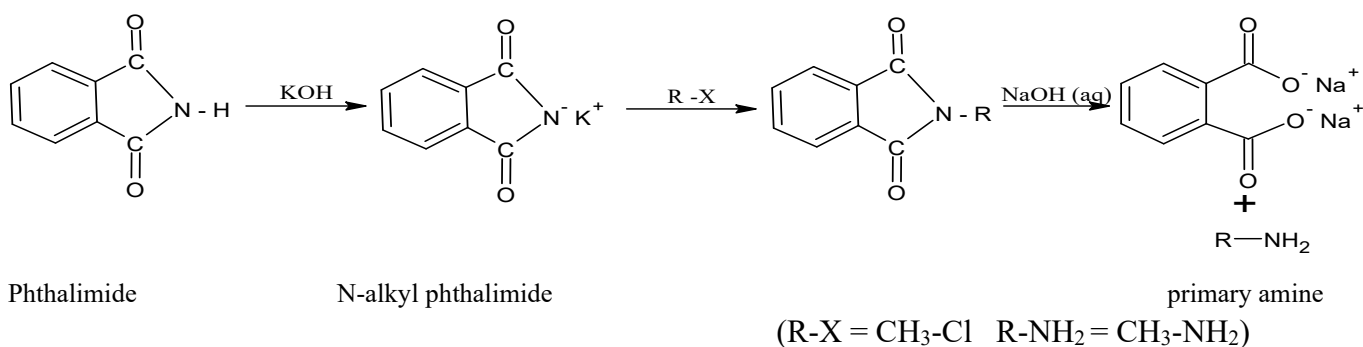
14. Discuss the coupling reaction of Benzene diazonium chloride with aniline. Give equation



Few most repeated questions:

1. Explain the preparation of primary amine by Gabriel phthalimide synthesis . Write the equation **OR**
 Give the equation for preparation of methyl amine (methanamine) by Gabriel phthalimide synthesis

Ans:



2. Between methanamine (CH₃NH₂) and aniline benzenamine (aniline)(C₆H₅NH₂) which is more basic?
 Give reason

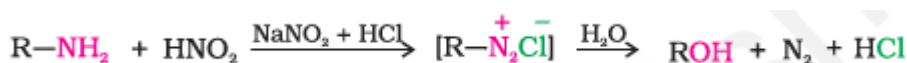
Ans: methanamine (CH₃NH₂), because +I (Positive inductive effect)

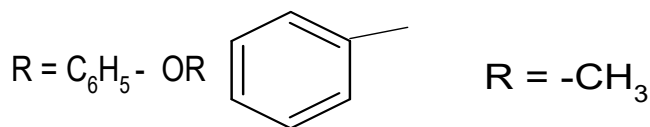
3. Give reason: aniline is a weaker base than ammonia **OR** Between ammonia and aniline, which is more basic?

Ans: Aniline, due resonance protonation of aniline decreases (less)

4. Name the major product formed write the equation when nitrous acid is treated with methylamine

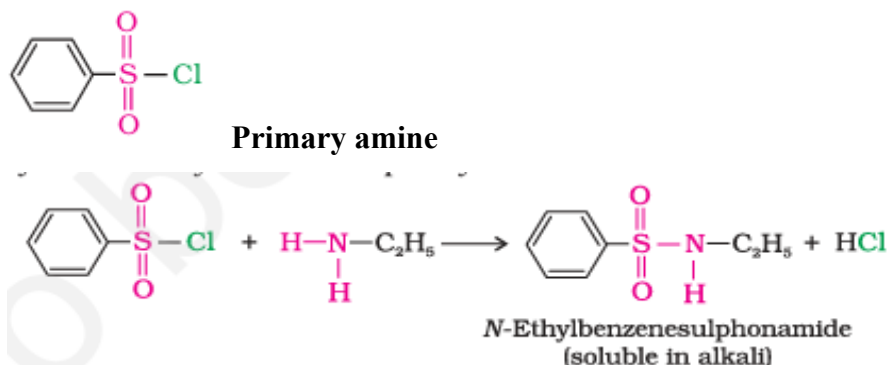
Ans:





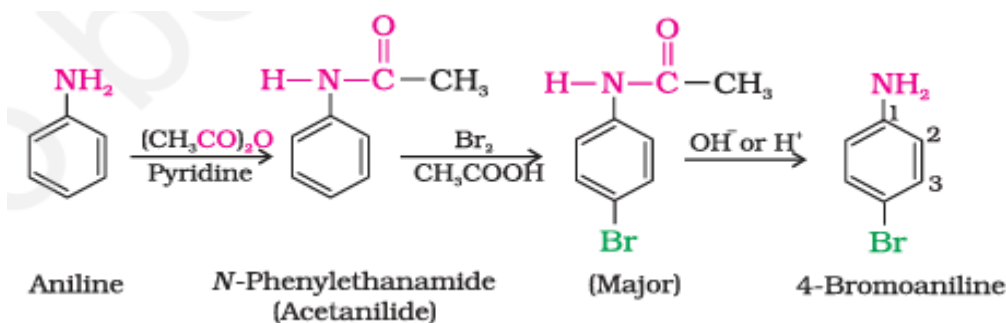
5. Write the structure of Hinsberg reagent. Which type of amines reacts with Hinsberg reagent to give precipitate, which is soluble in NaOH? Write its chemical reaction

Ans:



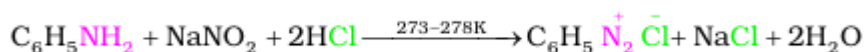
6. Explain with a chemical equation for the conversion of aniline to 4-bromoaniline.

Ans:



7. Explain the conversion of aniline to benzene diazonium chloride with chemical equation.

Ans:



Aniline Sodium Nitrate Benzene diazonium chloride

8. Explain the coupling reaction of Benzene diazonium chloride with phenol. Write the equation.

